Unfurling Light Sails For Advanced Deployment Technology:

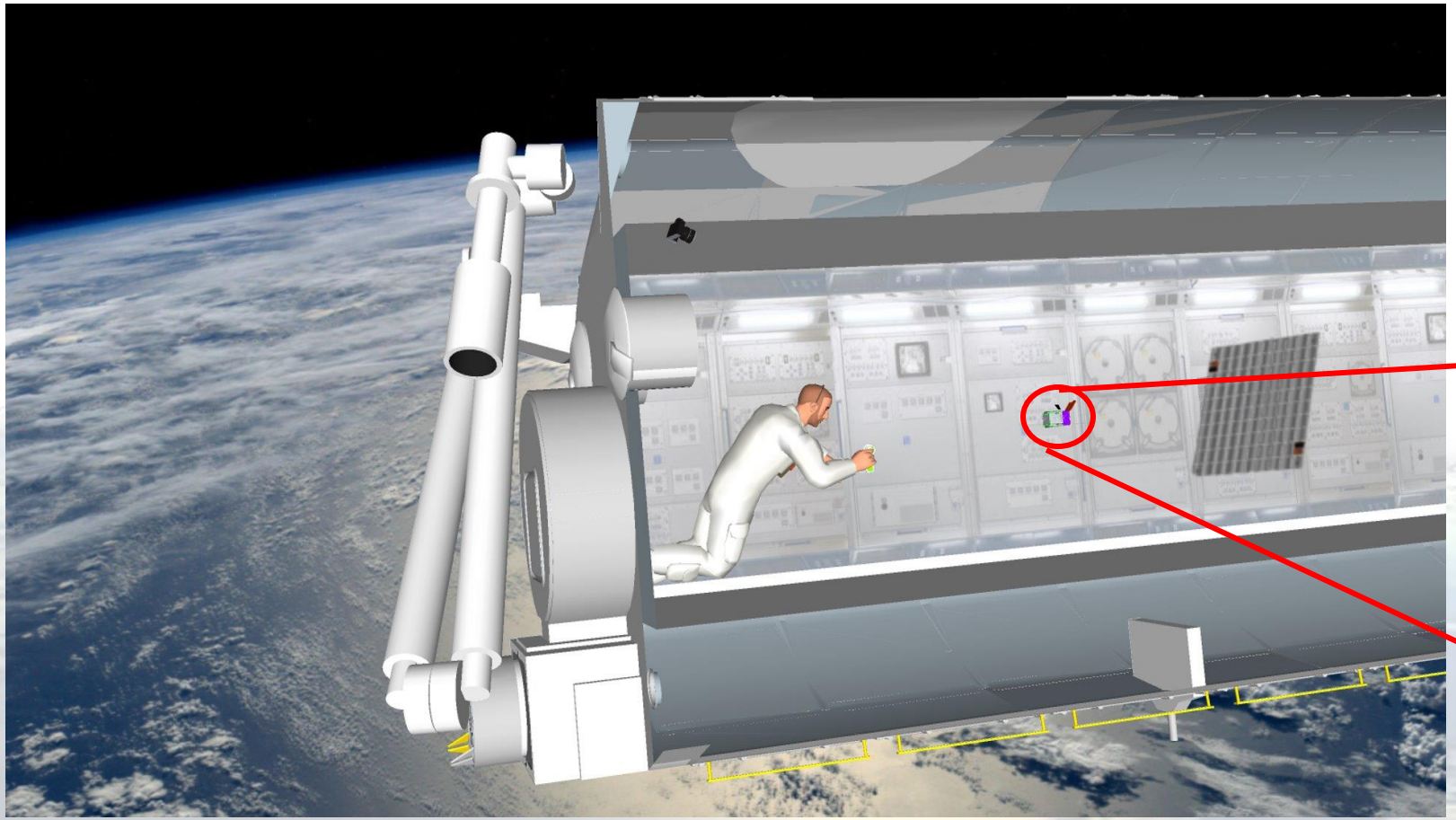
Sailing to the Stars Mission Mechanical Design

Claire Kim, Ethan Francolla, Gabriella Elcsics, Joshua Umansky-Castro

7th International Symposium on Space Sailing

Cornell University
Space Systems Design Studio
Sailing to the Stars

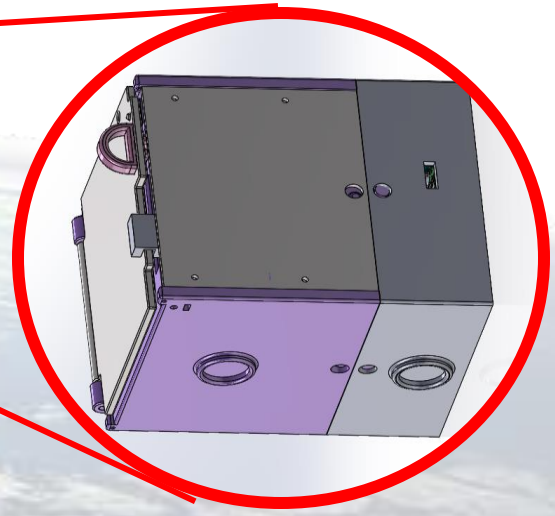
Sailing to the Stars



Sailing to the Stars:

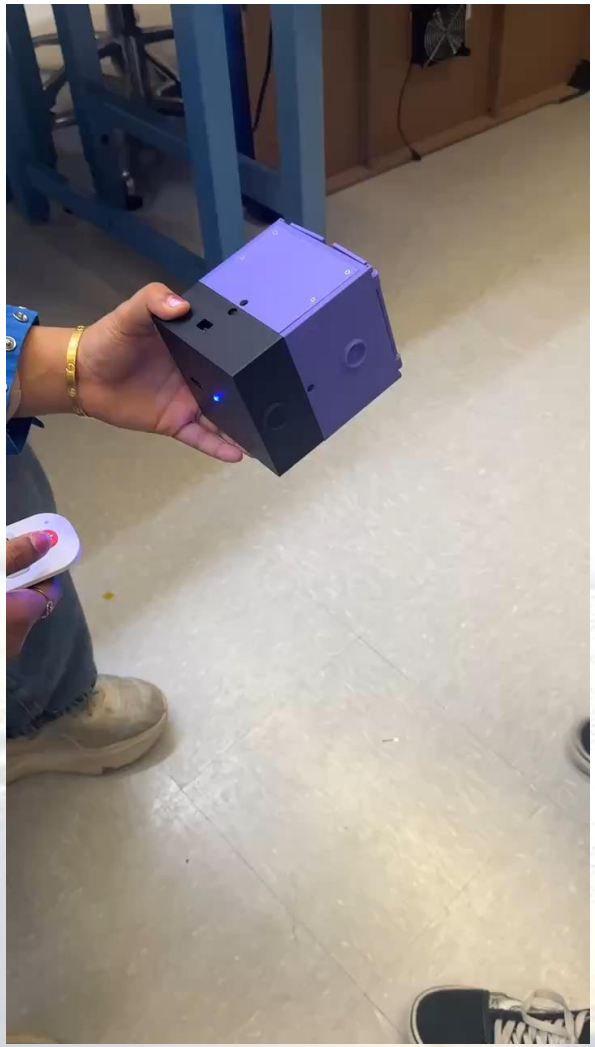
Light Sails in Microgravity

- *ISS Flight Experiment*
- *6 deployments, 60 minutes*



Flight Unit Deployment Tests

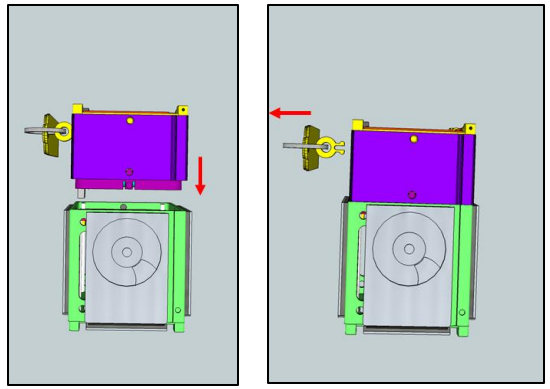
"1g" Test Environment of Light Sail





Successful deployments demonstrations in the lab

CONOPS



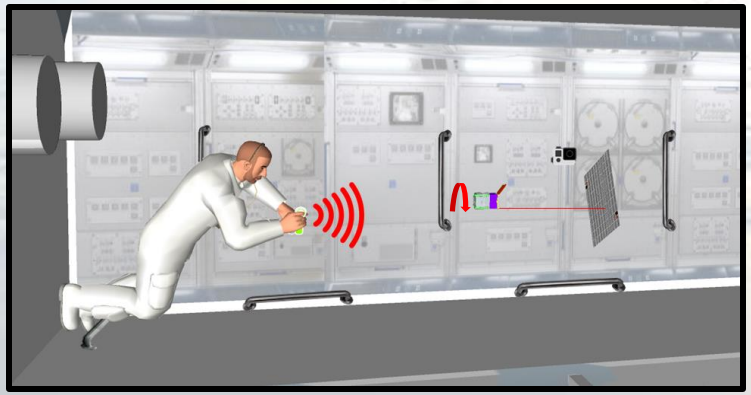
1) Mate Avionics Cube and Payload, and remove safety tag.



2) Gently release assembly with guide light aimed down the module.



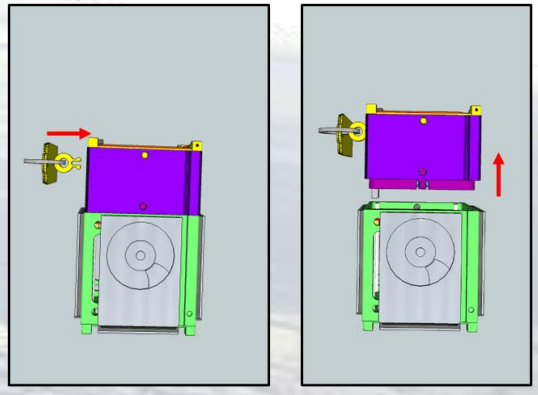
3) Use IR remote to activate deployer spin-stabilization sequence.



4) Use IR remote to trigger sail deployment.

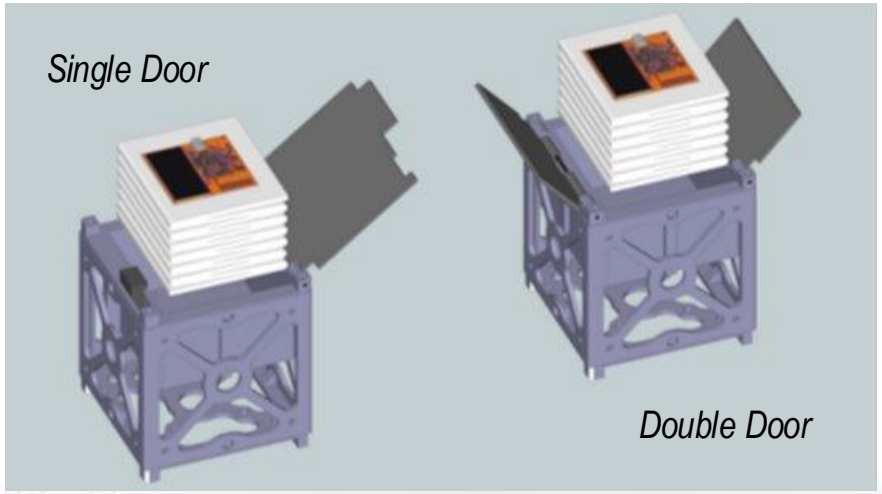


5) Use IR remote to spin down deployer.



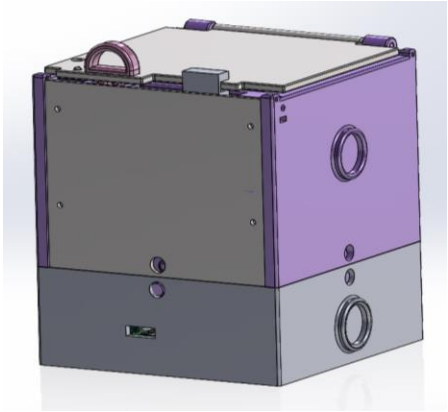
6) Reinsert safety tag and disconnect Cube and Payload.

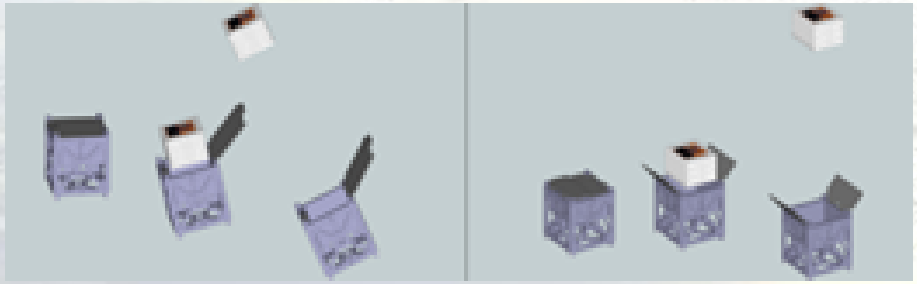
Mission Overview

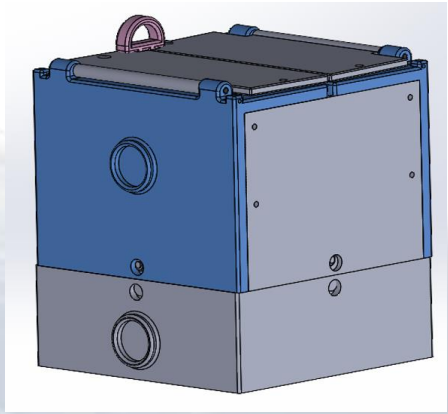


Project Goals:

1.	*Repeatable*	*Multiple plug and play units*
2.	*Transportable*	*~1U CubeSat form factor*
3.	*Reliable*	*Rigorous testing*
4.	*Easy to assembly*	*Fastener-less/Modular design*







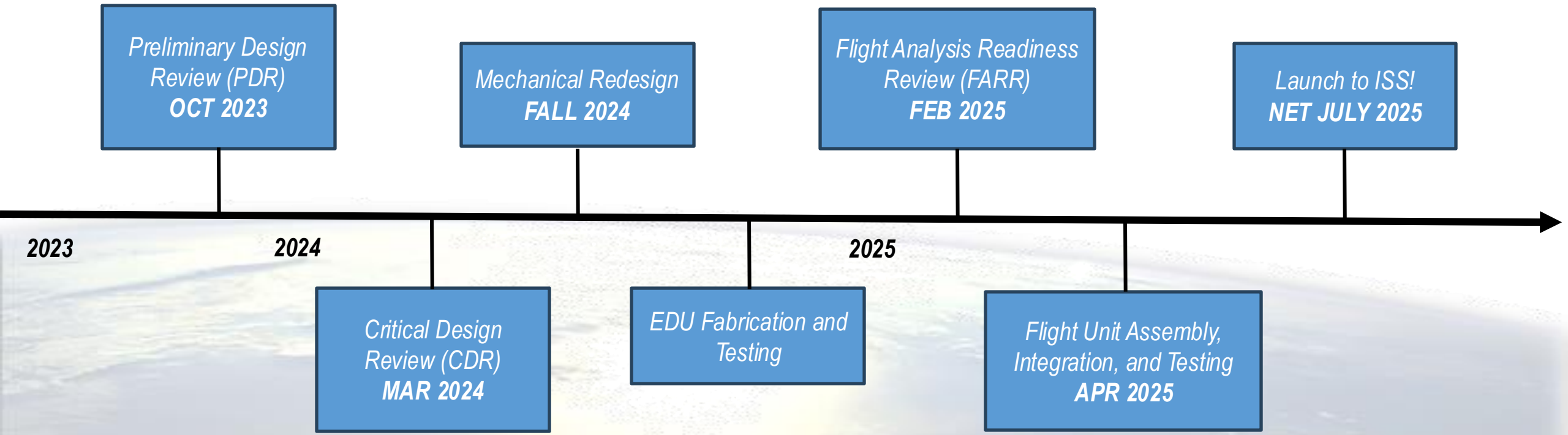
Anticipated "tip off" Behavior

Flight Units Ready for Transport

Single and Double-door Deployers

CAD Models Courtesy of Andy Filo

Project Timeline

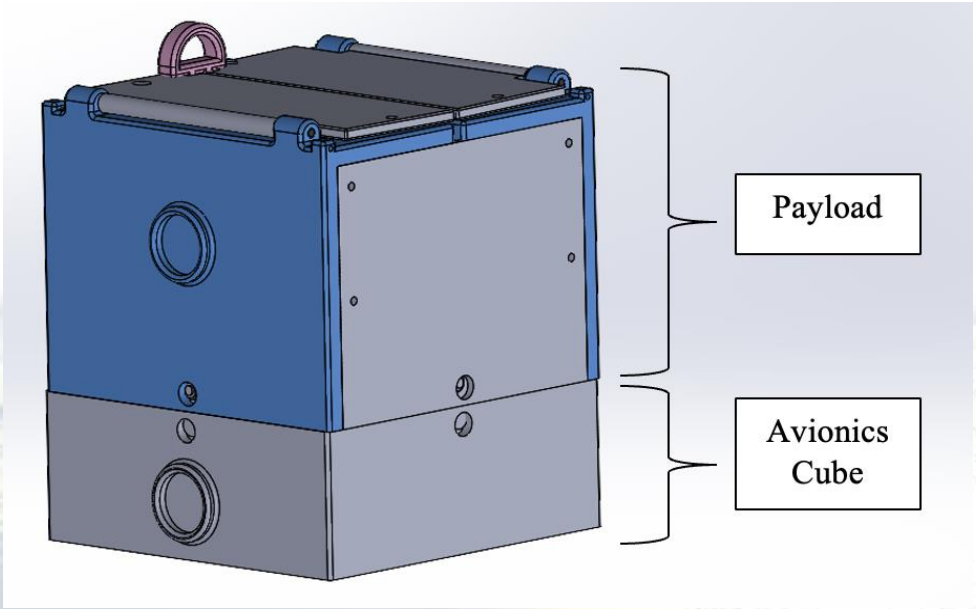


Mechanical Design

Deployer Design Overview

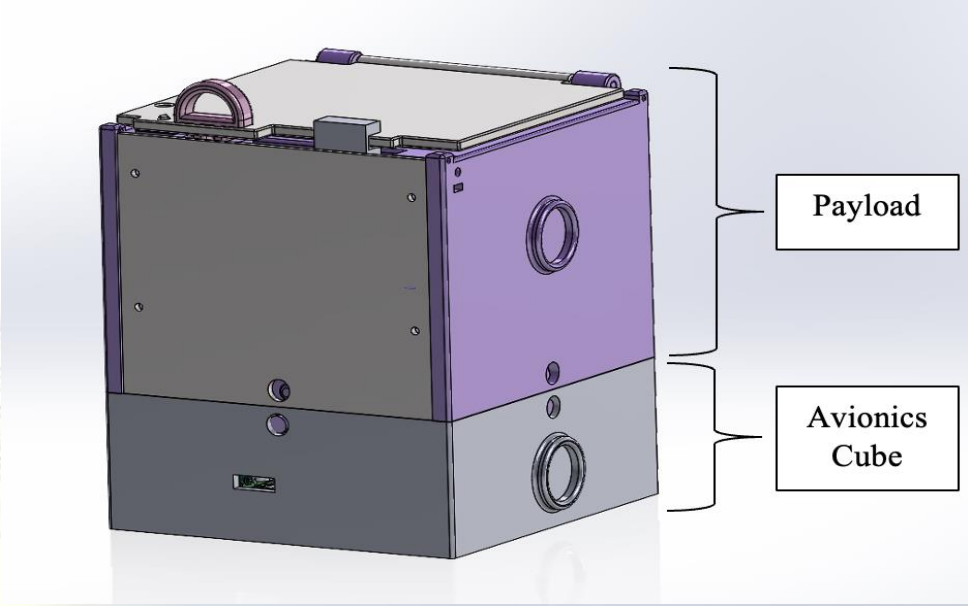


Double Door





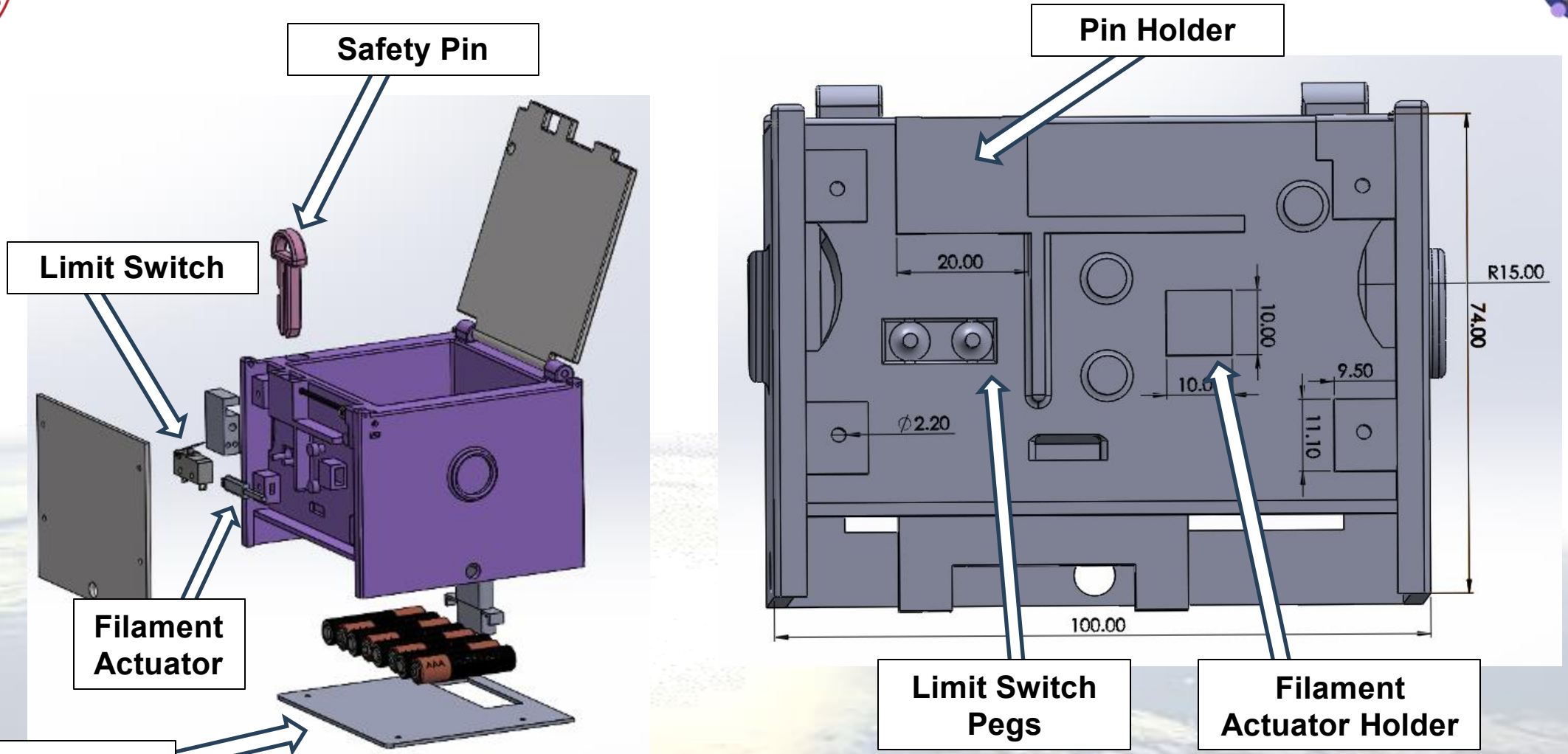
Single Door



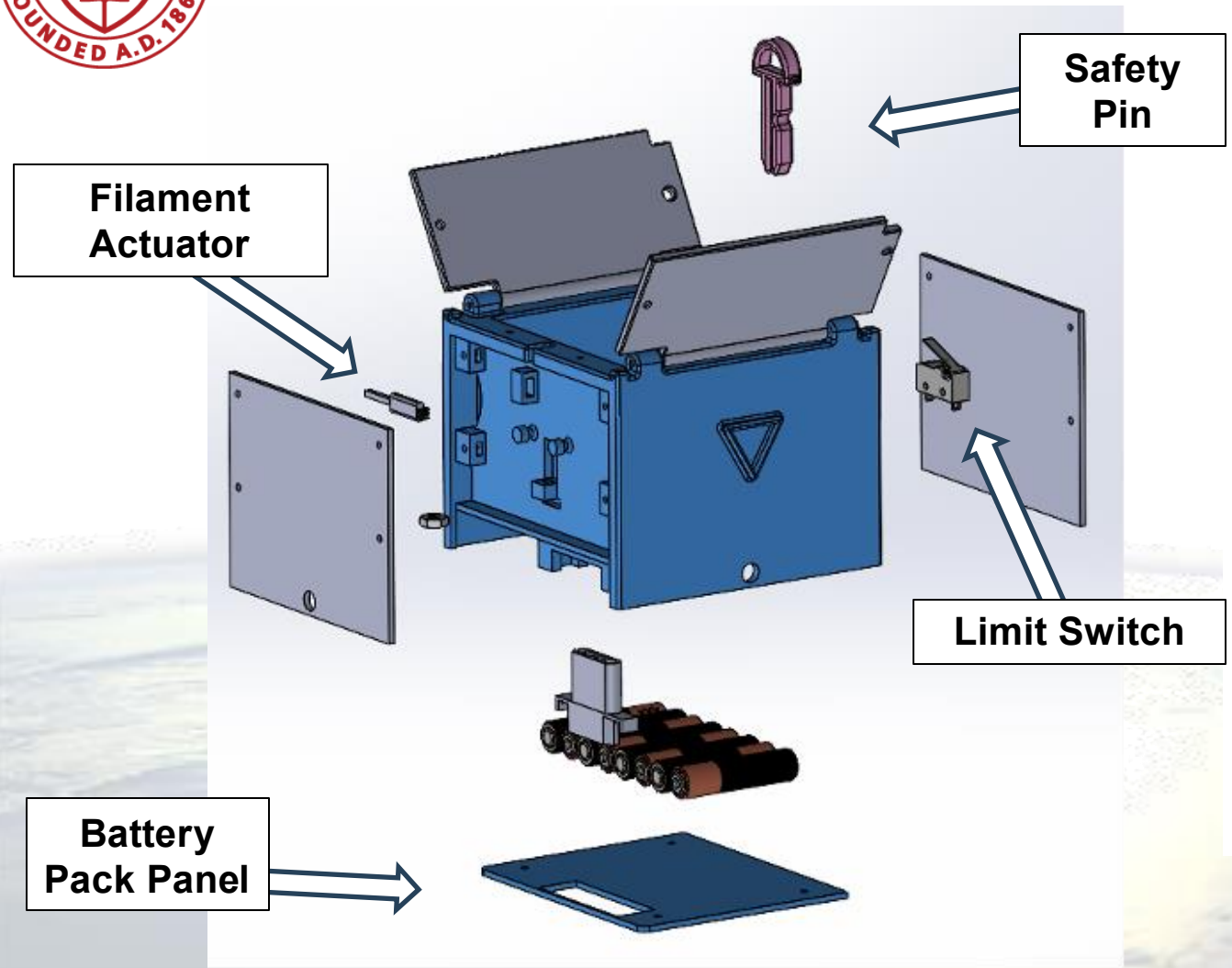
Modular Connection

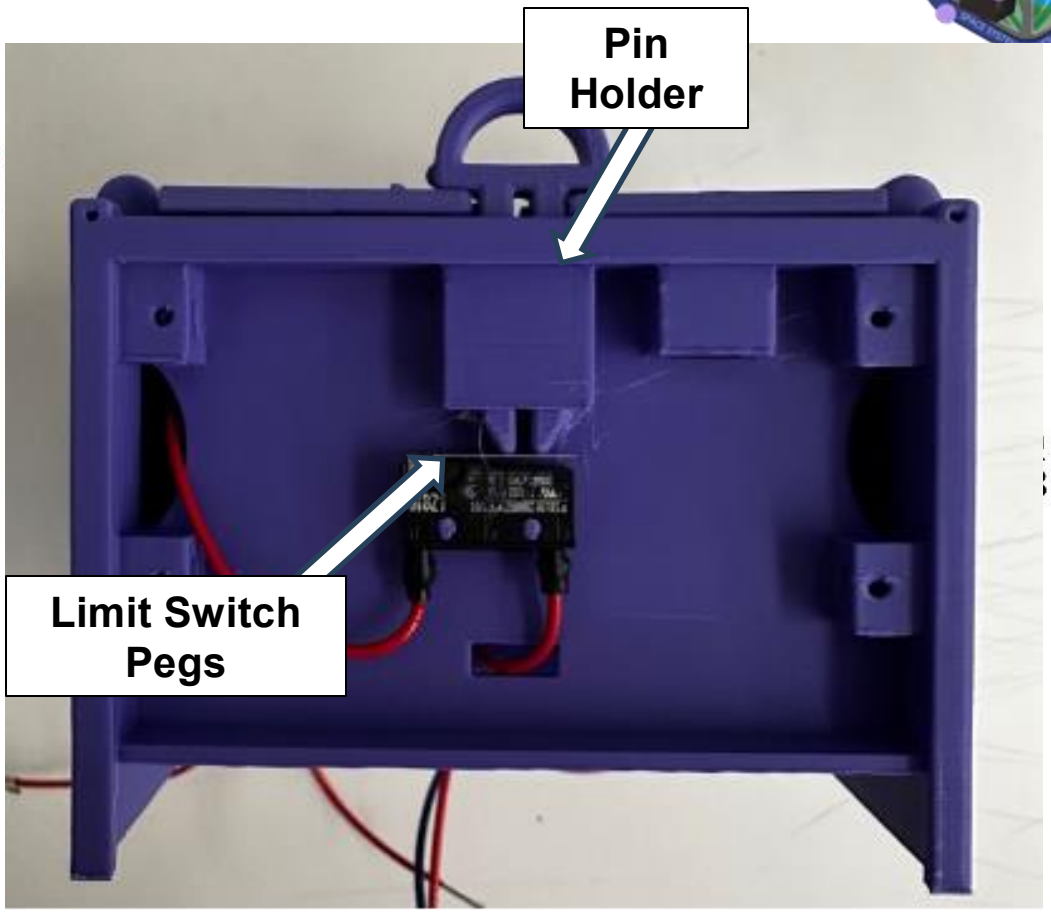


Single Door Payload Section: Exploded View

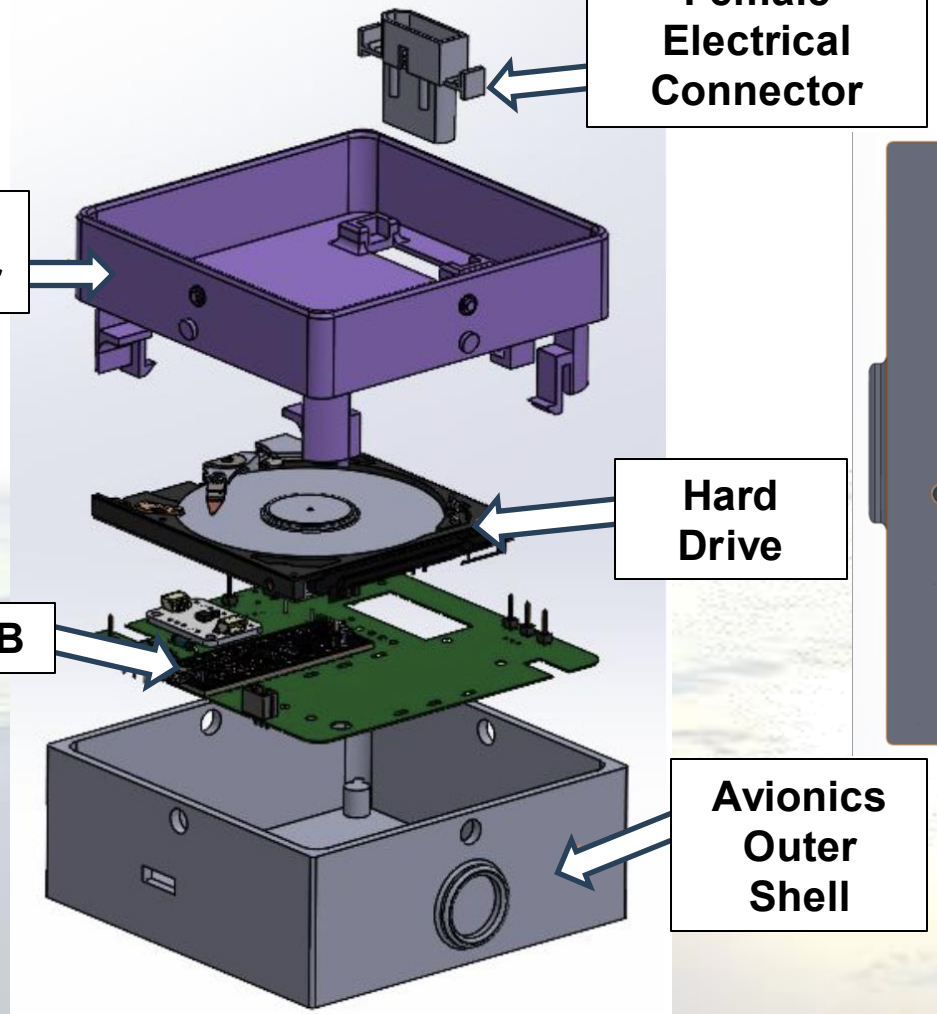


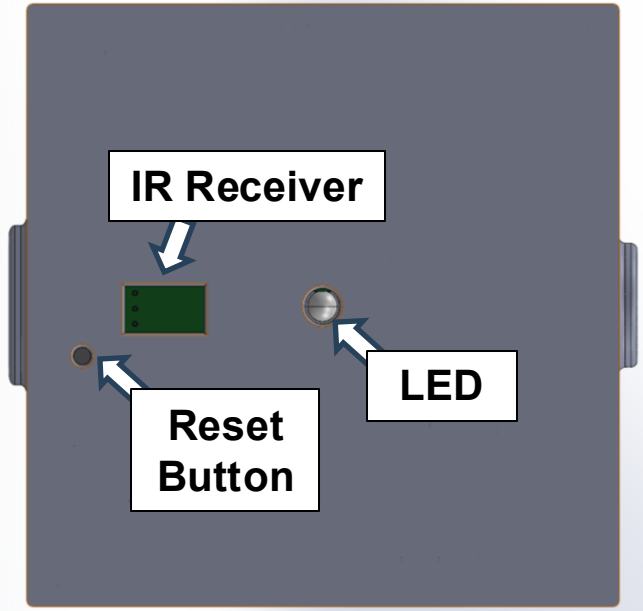
Double Door Payload Section: Exploded View

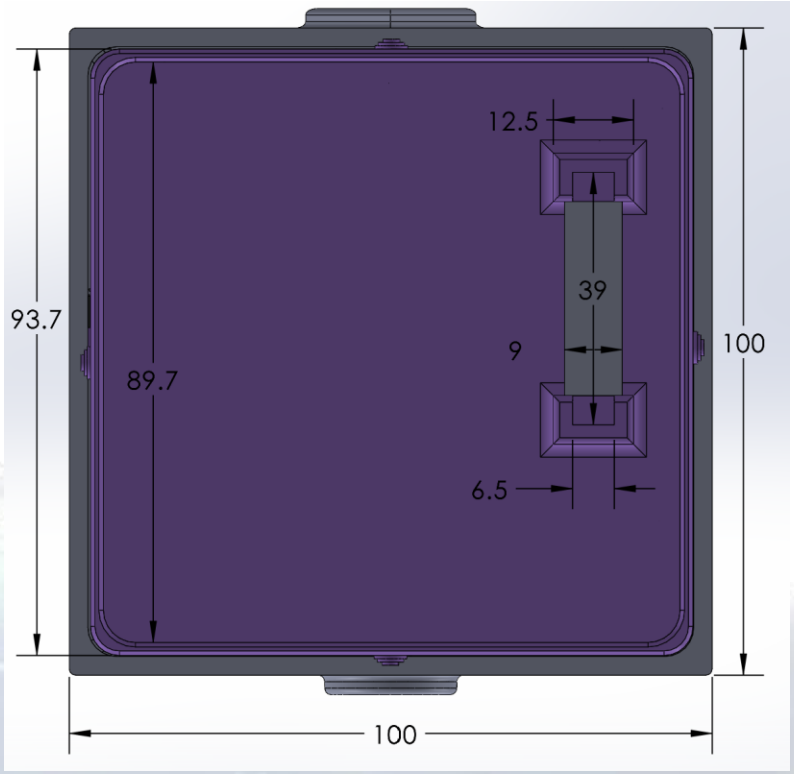




Avionics Section: Exploded View





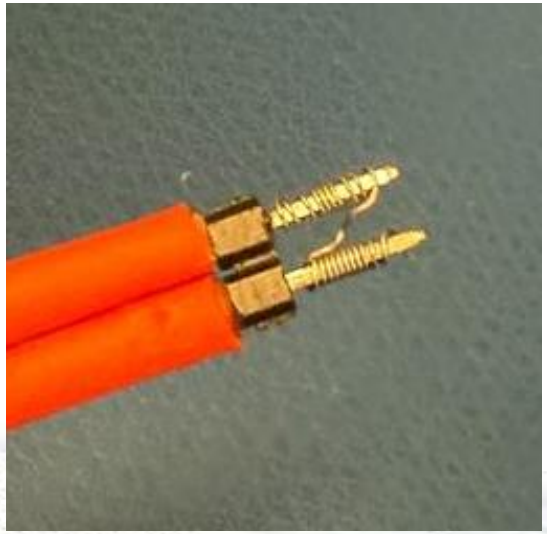


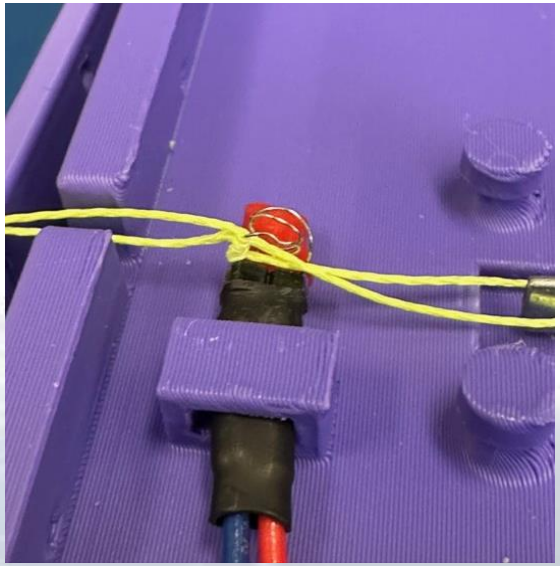
Filament Actuator

- *When activated, cuts the fishing line, allowing the door to open*



Single Door and Double Door Implementations

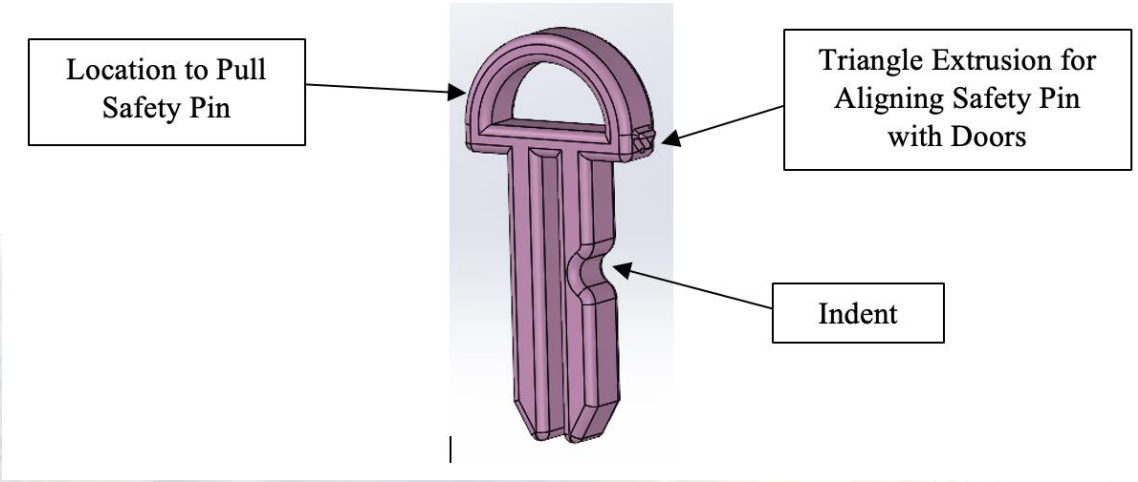


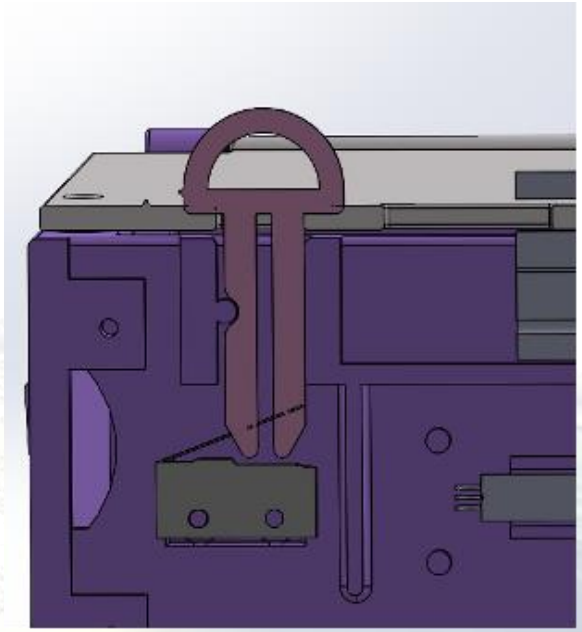


Single Door and Double Door Filament Actuators

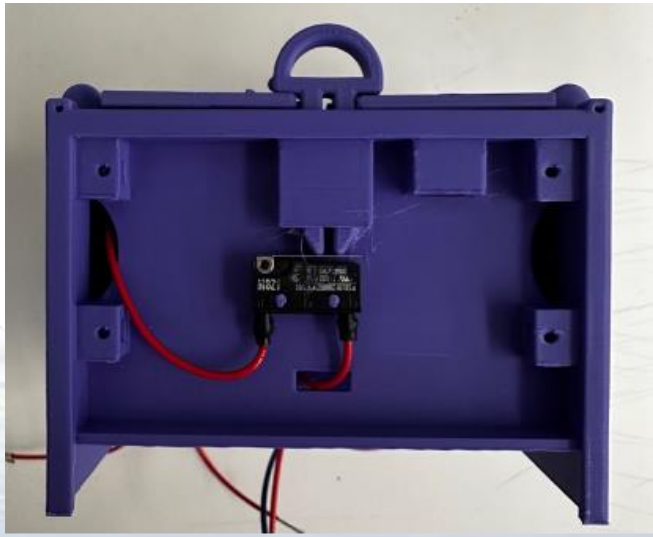
Safety Pin

- *Ensures secure door closure and prevents accidental power operation*





Cross-Section

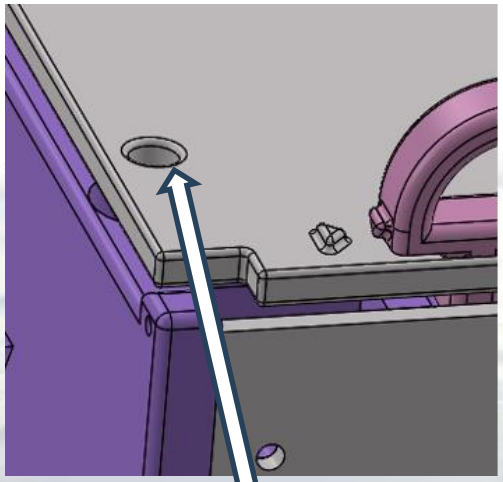


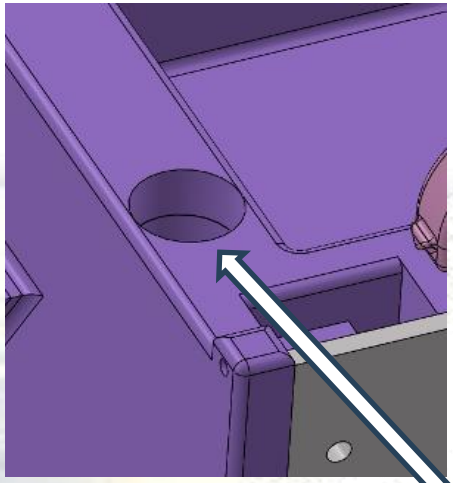
Limit Switch Depressed by Safety Pin

Guide Laser

- **Guide Laser Location:** *ensures proper deployer orientation*
- *Deployer designs feature a press-fit location for the 9 mm diameter light guide*

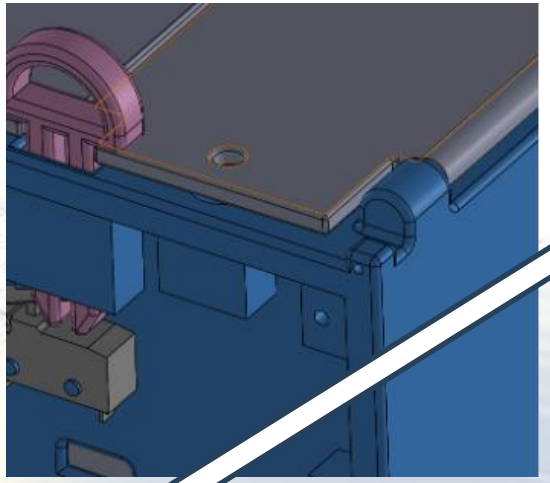
Single Door

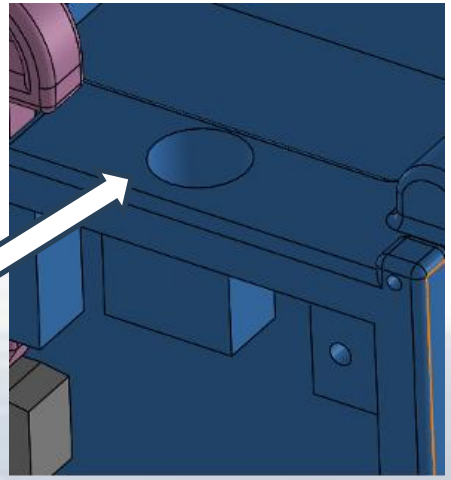




Hole for laser through door

Double Door





Press fit through hole in chassis

Assembly and Testing

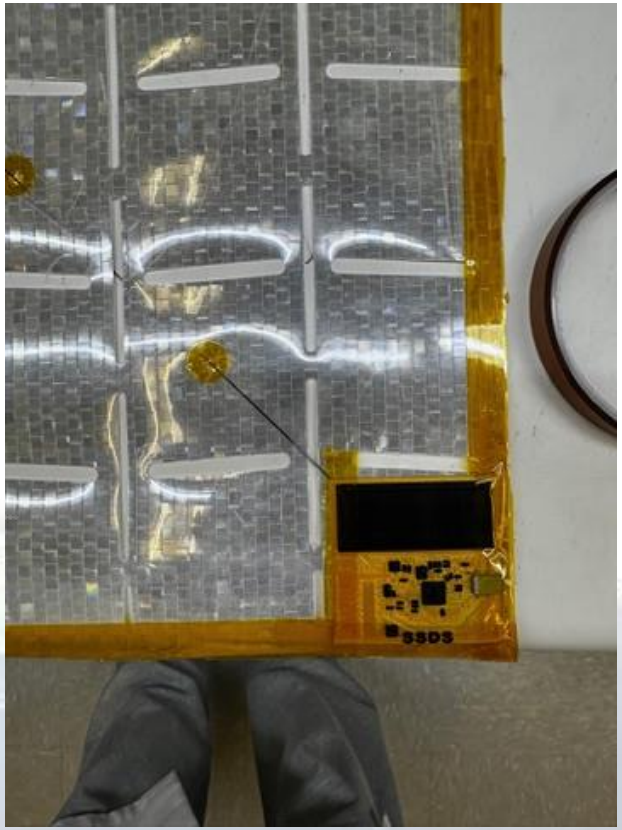
Light Sail Assembly



Sail during taping



Sails Ready for Epoxy

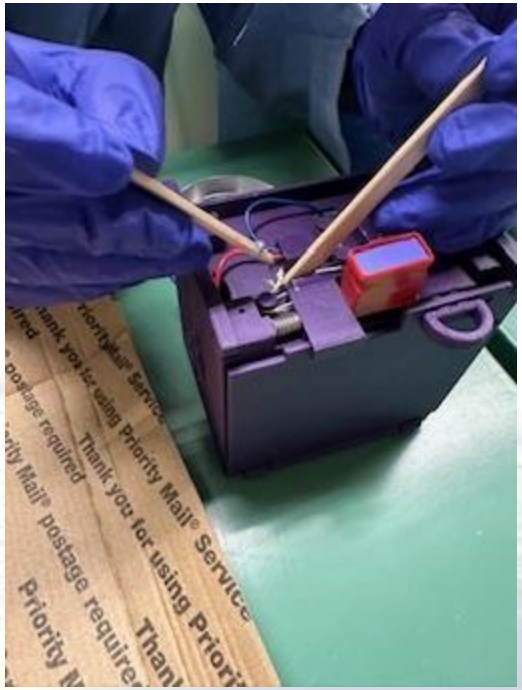


Integrated mock ChipSat

Flight Unit Assembly



Epoxying Guide Laser



Epoxying Fishing Line



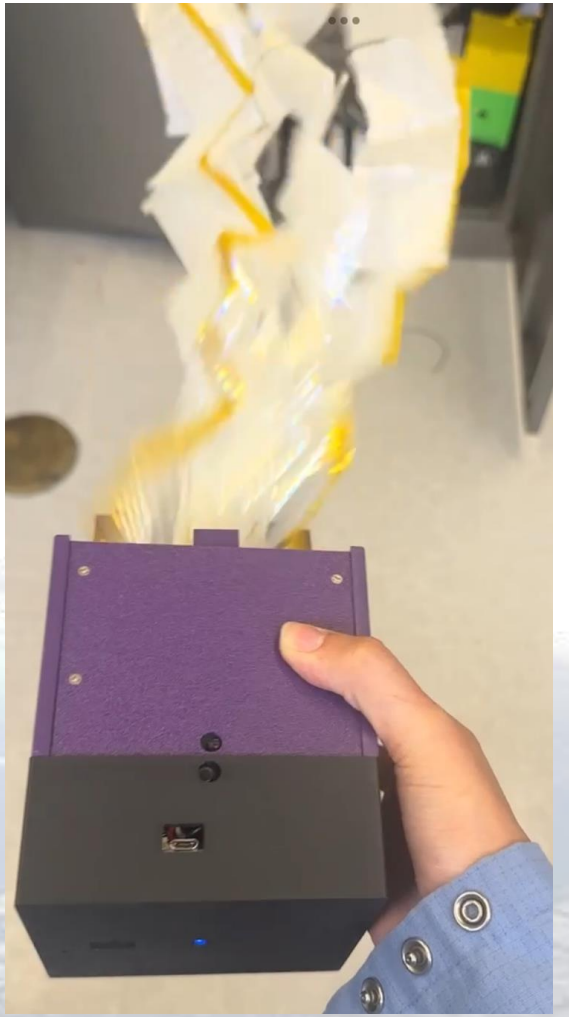
Inserted Light Sail

Battery Pack

1g Deployment Testing

- *Successfully tested all flight units (post-EDU testing)*

Unit	Test Date	Test Results	Notes
Flight SD 1	4/29	Successful	Mechanical deployment successful.
Flight SD 2	4/29	Successful	Mechanical deployment successful (2nd video recorded).
Flight SD 3	4/28	Successful	New door integrated and tested successfully.
Flight SD 4	4/27	Successful	- Ensured fishing line tension during epoxy application. - Automated sequence worked; IR remote failed. - Guide laser turned on when safety pin was removed. - Candle smoke smell noticed after deployment. - Ready for epoxy (non-mechanical issue).
Flight DD 1	4/29	Successful	- IR remote worked. - No issues with fishing line.
Flight DD 2	4/30	Successful	- Deployment successful - Deployment successful.
Flight DD 3	4/28	Successful	Test conducted without the light sail; fishing line was successfully cut.
Flight DD 4	4/30	Successful	- Fishing line tied incorrectly; only one side cut. - Coil bent significantly during deployment.



Completed Flight Units

Assembled Flight Units and Backups

Flight Units Stored for Transport

Next Steps

- *Launch to the ISS in late July 2025!*
  - *Conduct experiment, receive and analyze results*
  - *SpaceX Crew-11*
- *Team will be on-call during deployments*
- *Flight units will be returned and recorded data analyzed*

Questions?